



FACULTY OF COMMERCE

BACHELOR OF APPLIED ACCOUNTING, AUDITING & INFORMATION SYSTEMS

1. PROGRAMME : Bachelor of Applied Accounting, Auditing & Information Systems

2. PRESENTED TO : Senate

3. PRESENTED BY : Faculty of Commerce

4. YEAR/LEVEL OF STUDY : 2

5. MODULE TITLE : Computer Networks

6. MODULE CODE : NET225

7. DURATION : 16 Weeks

8. LECTURE HOURS PER WEEK: 2

9. TUTORIAL HOURS PER WEEK: 1

10. STUDENT INDEPENDENT LEARNING HOURS PER WEEK: 4

11. TOTAL MODULE CREDITS: 12

12. PREREQUISITE MODULES: None

13. COREQUISITE MODULES: None

14. DELIVERY METHODS:

14.1 Mode of Delivery : Face to Face

14.2 Teaching Methods : Lectures, demonstration and tutorials

14.3 Assessment Methods : Continuous assessment

: Mid and Final examinations

14.4 Assessment Weighting : Continuous assessment 10%

: Mid Semester examination 30%

: Final examination 60%

15. AIM(S) OF THE MODULE:

The aim of this module is to provide students with an exposure to the nature, purpose and methodology of organization network infrastructure.

16. LEARNING OUTCOMES:

On completion of this Module students should be able to:

a) Outline uses of computer networks

- b) Discuss different network types and technologies
- c) Analyse network protocols
- d) Discuss data presentation issues
- e) Describe the emerging technologies
- f) Explain Network security and cryptography

17. TOPICS OF STUDY:

17.1 Uses of Computer Networks:

- a) Business applications
- b) Home applications
- c) Mobile applications
- d) Social applications

17.2 Telecommunication systems:

- a) Components and functions of telecommunication systems b)
- Communication networks:
- c) Network hardware
- d) Network topologies,
- e) Private branch exchange,

17.3 Network Categories:

- a) Personal Area Networks (PAN)
- b) Local Area Network (LAN)
- c) Metropolitan Area Network (MAN)
- d) Wide Area Networks (WAN)
- e) Intranets and Extranets
- f) Virtual Private Networks (VPN)
- g) Internetworks
- h) Practical (setting up a network)

17.4 Wired and wireless networks:

- a) Wired communication media
- b) The electronic spectrum
- c) Types of wireless communications media
- d) Practical

17.5 Network technologies:

- a) Network services and Broadband technologies
- b) Value Added Networks
- c) Packet switching and Circuit switching
- d) Frame relay, asynchronous transfer mode
- e) Integrated service digital network DSL
- f) Cable modem and T1 line
- g) Convergence network and unified messaging

17.6 Layered architecture and reference models:

- a) Functions of the layers
- b) Open system Interconnection Model
- c) TCP/IP Protocol suite
- d) Services of TCP/IP layers to layers above them
- e) Comparison of models

17.7 Network errors:

- a) Framing
- b) Error and Flow control
- c) Error detection and correction (Error detection and correction codes)

17.8 Emerging Technologies:

- a) Communication over the internet
- b) Instant messaging
- c) Mobile networking
- d) Cloud computing

17.9 The Internet:

- a) The internet
- b) Narrowband, broadband and access providers
- c) The world wide web

17.10 Network and web security:

- a) Cyber and communication networks threats
- b) Introduction to Cryptography
- c) Symmetric-key and public-key algorithms
- d) Digital signatures and management of public keys
- e) Communication security
- f) Social issues

18. PRESCRIBED TEXTS:

Tanenbaum A, Wetherall D. J. (2011). Computer Networks. 5th Edition, Prentice Hall; ISBN-13: 978-0-13-212695-3.

19. RECOMMENDED TEXTS/READINGS:

Loudon K., Karol T. (2011) Management Information System: Managing the digital firm. 12th Edition

Kurose J. F, Ross K.W. (2013), Computer Networking, A Top Down Approach, 6th Edition, Pearson Education, Inc, ISBN-13: 978-0-13-285620-1

Williams, B., Sawyer, S., (2014) *Using Information Technology*, 11th Edition, McGraw-Hill Education; 11th Edition, ISBN-13: 978-0073516882